

Here is a detailed breakdown of Lecture 1: Introduction to Computer Graphics, based on the provided document.

1. Fundamental Definitions and Concepts

The lecture begins by distinguishing between computer graphics and related visual computing fields:

[cite_start]Computer Vision: This is the process of discovering what is present in the world and where it is by looking[cite: 39]. [cite_start]It is defined mathematically as

$\text{Vision} = \text{geometry} + \text{Measurement} + \text{interpretation}$ [cite: 47].

[cite_start]Computer Graphics: This field deals with the generation of 2D images from a 3D world represented in a computer, or from non-pictorial information[cite: 68, 73]. [cite_start]Its main tasks include modeling (creating the shape/geometry), rendering (generating the 2D image using light and perspective), and animation (describing movement over time)[cite: 74-77].

[cite_start]Image Processing: This deals with problems where both the input and the output are images, manipulating and analyzing them via a computer [cite: 70-71].

[cite_start]Pattern Recognition: This starts with a picture and transforms it into an abstract description, such as a set of numbers or symbols [cite: 68-69].

[cite_start]Graphics systems operate in two main modes: Passive Graphics, where application data is

simply mapped onto a display, and Interactive Graphics, where a user can interact with the image using

input devices like a mouse [cite: 91-94].

[cite_start]When dealing with digital image formats, two of the most important parameters to keep in mind are spatial resolution and color encoding[cite: 116]. Different formats handle these parameters and data compression in ways that make them suitable for specific tasks.

GIF (Graphics Interchange Format)

- [cite_start]**Best Used For:** Text, line drawings, screenshots, cartoons, and simple web animations[cite: 128, 131, 132].
- [cite_start]**Color Limitations:** It is limited to a maximum of 256 colors[cite: 129].
- [cite_start]**Compression:** Uses the LZW compression algorithm, which provides lossless data compression that can reduce the original file size by 20-25% without losing quality[cite: 134].
- [cite_start]**Pros:** Small file sizes that are fast to load, widely supported by web browsers, and supports both transparencies and animations[cite: 130, 135, 136].
- [cite_start]**Cons:** Because of the 256-color limit, complex pictures and photographs look terrible, and fine details are lost[cite: 137].

JPEG / JPG (Joint Photographic Experts Group)

- [cite_start]**Best Used For:** Color photographs and millions-of-color images[cite: 139, 146]. [cite_start]It became incredibly popular alongside the rise of digital cameras[cite: 141, 142].
- [cite_start]**Compression:** Uses lossy compression, meaning visual quality is intentionally discarded to save space, and this lost data cannot be restored[cite: 144].
- [cite_start]**Pros:** Achieves very small file sizes while supporting millions of colors, making it perfect for sharing photos over the internet[cite: 146].
- [cite_start]**Cons:** It handles sharp edges poorly, tending to blur them[cite: 140, 145]. [cite_start]Furthermore, every time you edit and save a JPEG, it loses more and more image quality due to the repeated compression[cite: 148].

PNG (Portable Network Graphics)

- [cite_start]**Best Used For:** Websites, logos, and screenshots[cite: 152].
- [cite_start]**Features:** Introduced in 1999, it was designed to improve upon and replace GIF in many web applications[cite: 150, 151].
- [cite_start]**Color Types:** Unlike JPEG (which doesn't do palettes) and GIF (which only does palettes), PNG supports three main image types: true color, grayscale, and palette-based (8-bit) [cite: 153].

TIFF (Tagged Image File Format)

- [cite_start]**Best Used For:** Scanning data, faxing, and word processing[cite: 158].
- [cite_start]**Features:** Currently owned by Adobe, this format hasn't been structurally changed since 1992[cite: 156]. [cite_start]It is unique because it can store the image data and metadata (tags) in a single file[cite: 157].
- [cite_start]**Color & Compression:** Supports a massive variety of color levels (binary, grayscale, palettes, RGB, CMYK) and multiple compression methods[cite: 159, 160].
- [cite_start]**Pros:** Perfect image quality with absolutely no loss[cite: 161].
- [cite_start]**Cons:** The resulting files are massive, making them difficult to transfer and completely unsuitable for standard web viewing without specialized programs[cite: 162, 163].

BMP (Bitmap)

- [cite_start]**Best Used For:** A foundational, device-independent format used frequently in Windows systems[cite: 168, 169].
- [cite_start]**Features:** It is based on the RGB color model and uses absolutely no compression[cite: 170, 171]. [cite_start]The file is split into a header region (containing size and color depth) and a data region (containing the literal value of every single pixel)[cite: 172, 173].
- [cite_start]**Pros:** Extremely high-quality images that are easy to edit and change with zero quality loss during processing[cite: 174].

- [cite_start]**Cons:** The lack of compression means very large file sizes, making them difficult to display efficiently on the internet[cite: 174].

Other Formats Discussed

- [cite_start]**PostScript & EPS:** PostScript is a programming language from 1985 optimized specifically for printing graphics and text[cite: 120, 121, 122]. [cite_start]Because standard PostScript needs to know the image size in advance, the EPS (Encapsulated PostScript) format was created to solve this by containing the image and comment lines[cite: 123, 124, 125].
- [cite_start]**XBM & XPM:** Formats used in the Unix world for background images and program icons[cite: 176]. [cite_start]They use no compression[cite: 177]. [cite_start]XBM is monochrome (storing one byte array per line), while XPM supports color[cite: 178, 179].

Here is a deeper dive into Digital Image Basics and Aliasing, expanding on the concepts from the first lecture.

1. Digital Image Basics

A digital image is fundamentally a grid of data. [cite_start]Mathematically, it is defined as a 2D light intensity function $f(x,y)$, where x and y denote spatial coordinates, and the value of f at any specific point represents the brightness or gray level of the image at that exact location [cite: 184-185]. [cite_start]This information is stored in a computer as a 2D matrix where each individual value corresponds to a single pixel [cite: 117, 186-193].

Image Types

Images are categorized by how much data each pixel holds:

- **Binary Images:** The simplest form, where a pixel is either black or white. [cite_start]This is represented as $g(x,y) \in \{0,1\}$ [cite: 260-261].
- **Gray-scale Images:** Pixels represent varying shades of gray, typically ranging from 0 (black) to 255 (white). [cite_start]This is represented as $g(x,y) \in \mathbb{C}$ [cite: 262-264].
- [cite_start]**Color Images:** These are composed of three separate channels (Red, Green, and Blue) overlapping each other, represented as $g_R(x,y) \in \mathbb{C}$, $g_G(x,y) \in \mathbb{C}$, and $g_B(x,y) \in \mathbb{C}$ [cite: 265-267].

Core Image Characteristics

- [cite_start]**Bit Depth:** This determines the amount of information each pixel holds, dictating how many colors or tones can be displayed[cite: 290, 293].
 - [cite_start]**1-bit** per pixel yields 2 colors (strictly black and white)[cite: 295].
 - [cite_start]**8-bits** per pixel yields 256 colors or gray tones[cite: 296].
 - [cite_start]**24-bits** per pixel yields 16,777,216 colors, providing smooth gradations comparable to traditional color film[cite: 297].

- [cite_start]*Note on Flexibility*: Direct coding allows different bit allocations per primary color; for instance, 10 bits per RGB channel yields over 1 billion colors ($2^{10} \times 2^{10} \times 2^{10}$) [cite: 298-300].
 - [cite_start]**Spatial Resolution**: This is the smallest discernible detail in an image, defined as the number of image points (pixels) per unit measure [cite: 323-324]. [cite_start]A higher sampling rate and larger grayscale improve the approximation of the digital image to the original scene [cite: 307].
 - [cite_start]**Image Storage Size**: The total number of bits required to store an uncompressed digitized image can be calculated using the formula $b = M \times N \times k$, where M and N are the image dimensions (width and height in pixels), and k is the number of bits per pixel [cite: 340-342].
 - [cite_start]**Aspect Ratio**: This is the proportional relationship between the width (x) and height (y) of an image or pixel, calculated as x/y [cite: 352-358].
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2. Aliasing

[cite_start]Aliasing is a visual distortion that occurs when a continuous image is not sampled densely enough during digitization [cite: 367]. [cite_start]Because the discrete pixel grid cannot perfectly capture high-frequency details (like sharp edges or the perspective projections of textures), artifacts appear in the final image [cite: 368-369].

[cite_start]To deal with aliasing, you can either pass the signal through a low-pass filter before sampling it, or you can increase the sampling frequency (resolution) [cite: 371-373].

Types of Aliasing

Aliasing manifests in two major ways within computer graphics:

A. [cite_start]**Spatial Aliasing (Problems in static images)** [cite: 378-379]

- **Jagged Profiles ("Jaggies")**: The most familiar effect, where smooth curves or diagonal lines appear as rough, zigzagged stair-steps. [cite_start]This is especially noticeable in areas with high contrast between the inside and outside of a silhouette [cite: 383-385].
- **Loss of Detail (Improperly rendered detail)**: When fine details fall between the gaps of the pixel grid, they can be rendered incorrectly. [cite_start]Small shapes might disappear entirely, double in width, or render at inconsistent sizes compared to identical neighboring shapes [cite: 390-392].
- **Disintegrating Textures**: When a textured surface (like a checkerboard floor) stretches into the distance, the pattern should naturally get smaller. [cite_start]Due to aliasing, the distant shapes may instead appear larger, distorted, or irregularly shaped because the pixel grid can no longer accurately capture the dense details [cite: 398-399].

B. [cite_start]**Temporal Aliasing (Problems in motion/video)** [cite: 380-381]

- **Strobing:** When motion pictures do not have a sufficiently high frame rate, the human brain stops interpreting the sequence as continuous motion, making the movement look jerky. [cite_start]This is common when viewing videos over slow internet connections [cite: 402-406].
- **Flickering:** This occurs when the frame update rate is too slow. [cite_start]For instance, on older CRT monitors, if the refresh rate is low, the screen brightness drops for a fraction of a second—just long enough for the human eye to notice the flash [cite: 408-411].

5. Applications of Computer Graphics Computer graphics are utilized across almost every major industry:

- Building Design and Construction:** Visualizing architectural structures and allowing clients to "walk through" buildings.
- Electronics Design:** Designing printed circuit boards (PCBs) on video screens before manufacturing.
- Mechanical Design:** Using constructive solid geometry (CSG) to test and design machine parts.
- Entertainment and Animation:** Making cartoon animation films.
- Aerospace Industry:** Simulating airflow and shock waves over aircraft wings to detect turbulence.
- Medical Technology:** Composing 3D models of the brain via CAT scans to detect tumors and studying drug effects on tissues.
- Cartography:** Producing accurate geographical, relief, and weather maps.
- Other Uses:** Art and commerce, presentation graphics, data visualization, video games, education/training, and designing graphical user interfaces (GUIs).